Simplicial nef and Mori cones of normalized Chow quotients via primal curves

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Editorial scope. Count one principal dual-cone theorem: Theorem 1.2 in its precise form Theorem 4.10. Lemma 4.2 supplies Picard-number and basis support within this unit. The boundary theorem is retained as support here and counted only once, in PM-0005. Do not count the primal-curve lemmas, intersection lemmas, partial-quotient theorem, canonical formula or examples separately.

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Allowed standard prerequisites (editorial).

- Białynicki-Birula local/decomposition theory and equivariant vectorbundle weights. Current§2.1 andBB73,Car02,Ive72; exact normalbundle/cell notation retained.. Established torus-action tools; none of current-specific boundary/primal/cone proofs is removed.
- Deformation theory of nodal curves, local-to-globalExt, Hilberttangent/obstruction identification. CurrentProposition3.7 proof andTal09§6,Sernesi2006; standard quoted statements retained exactly.. Classical deformation-theoretic inputs; current Kodaira–Spencer surjectivity, torusweight-zero passage and boundary identification are full author text.
- Numerical divisor/curve duality, nef pullbacks, projectivebundle/blowup Picard behavior, Mori bend-and-break. Current§4; earlier separately licensed amplenessproof refers DebarreProposition3.2.. Established general geometry; current primal construction and its full localconic/globalcontraction argument are not prerequisites.

2 Background material

Throughout the paper, we will work over the field of complex numbers. The term *rational homogeneous variety* will always refers to a projective variety that supports a transitive algebraic action of a semisimple group. When working with these varieties, we will use the notation of marked Dynkin diagrams, as described, for instance, in [ORSW23a, Section 3.1].

2.1 $\mathbb{C}^*$-actions

In this section, we present some background material on $\mathbb{C}^*$-actions. We refer to [BB73, Car02, ORSW23b] for further details. In what follows, X will be a smooth projective variety, endowed with a nontrivial $\mathbb{C}^*$-action.

We denote by $X^{\mathbb{C}^*}$ the fixed locus of the action, and by $\mathcal{Y}$ the set of irreducible fixed-point components:

$$X^{\mathbb{C}^*} = \bigsqcup_{Y \in \mathcal{Y}} Y.$$

The *sink* and the *source* of the action, often called the *extremal fixed-point components*, are the fixed-point components containing, respectively, the limiting points

$$x_- := \lim_{t \to 0} t^{-1}x, \quad x_+ := \lim_{t \to 0} tx$$

of a general $x \in X$. The fixed-point components different from the sink and the source are called *inner*. Every $Y \in \mathcal{Y}$ is smooth (*cf.* [Ive72]), and the normal bundle of Y in X splits into two subbundles, on which $\mathbb{C}^*$ acts with positive and negative weights, respectively:

$$N_{Y|X} \simeq N^+(Y) \oplus N^-(Y). \tag{2.1}$$

We set $\nu^\pm(Y) := \mathrm{rk}(N^\pm(Y))$.

2.1.1 Białynicki-Birula decomposition

Given any set $S \subset X$, we define

$$X^\pm(S) := \{x \in X \mid x_\pm \in S\}, \quad B^\pm(S) := \overline{X^\pm(S)}. \tag{2.2}$$

When $S = Y$ with $Y \in \mathcal{Y}$, we call $X^\pm(Y)$ the positive and the negative *Białynicki-Birula cells* of the action at Y (BB-cells, for short); we have decompositions

$$X = \bigsqcup_{Y \in \mathcal{Y}} X^+(Y) = \bigsqcup_{Y \in \mathcal{Y}} X^-(Y).$$

We will denote by $B^\pm(Y)$ the closures of the BB-cells $X^\pm(Y)$. Every cell $X^\pm(Y)$ has a natural morphism onto Y, sending every $x \in X^\pm(Y)$ to its limiting point $x_\pm$. It was shown by Białynicki-Birula that $X^\pm(Y)$ is an affine bundle, locally isomorphic to $N^\pm(Y)$.

2.1.2 Actions on polarized pairs

Given a line bundle $L \in \mathrm{Pic}(X)$, there exists a linearization of the $\mathbb{C}^*$-action on it, so that for every $Y \in \mathcal{Y}$, $\mathbb{C}^*$ acts on $L_{|Y}$ by multiplication with a character $m \in \mathrm{M}(\mathbb{C}^*) = \mathrm{Hom}(\mathbb{C}^*, \mathbb{C}^*)$, called the *weight of the linearization on* Y. Two linearizations differ by a character of $\mathbb{C}^*$. In particular, for any $L \in \mathrm{Pic}(X)$, there exists a unique linearization (called *normalized linearization*) whose weight at the sink is equal to zero. Fix an isomorphism $\mathrm{M}(\mathbb{C}^*) \simeq \mathbb{Z}$; then this linearization defines a map $\mu_L \colon \mathcal{Y} \to \mathbb{Z}$, called the *weight map*.

A $\mathbb{C}^*$*-action on the polarized pair* (X, L) is a $\mathbb{C}^*$-action on X, together with the weight map μ_L determined by an ample line bundle L. In this case, we denote by

$$0 = a_0 < \cdots < a_r$$

the weights $\mu_L(Y)$, $Y \in \mathcal{Y}$, and set

$$Y_i := \bigcup_{\mu_L(Y) = a_i} Y.$$

The values a_i will be called the *critical values*, and the number r will be called the *criticality* of the action. The minimum and maximum of the critical values are achieved only at the sink and the source, which are then equal to Y_0 and Y_r, respectively. The value $\delta = a_r$ is called the *bandwidth* of the $\mathbb{C}^*$-action on (X, L).

2.1.3 Equalized actions

A $\mathbb{C}^*$-action is *equalized* at $Y \in \mathcal{Y}$ if for every $x \in (X^-(Y) \cup X^+(Y)) \setminus Y$, the isotropy group of the action at x is trivial. Equivalently (see [ORSW22, Lemma 2.1]), $\mathbb{C}^*$ acts on $N^+(Y)$ with all the weights equal to $+1$ and on $N^-(Y)$ with all the weights equal to -1. The action is called *equalized* if it is equalized at every $Y \in \mathcal{Y}$.

For an equalized action, the closure of any 1-dimensional orbit is a smooth rational curve, whose L-degree may be computed in terms of the weights at its extremal points (*cf.* [ORSW22, Lemma 2.2], [RW22, Corollary 3.2]).

Lemma 2.1 (AM vs. FM)**.** *Let (X, L) be a polarized pair with an equalized $\mathbb{C}^*$-action, and let C be the closure of a 1-dimensional orbit, with sink x_- and source x_+. Then C is a smooth rational curve of L-degree equal to $\mu_L(x_+) - \mu_L(x_-)$.*

Using Lemma 2.1, one can show (see [OS26, Corollary 2.4]) the following.

Corollary 2.2. *Let X be a smooth projective variety with an equalized $\mathbb{C}^*$-action, and let C be the closure of a 1-dimensional orbit, with sink in Y and source in Y'. Then*

$$-K_X \cdot C = \left(\nu^+(Y') - \nu^-(Y')\right) - \left(\nu^+(Y) - \nu^-(Y)\right).$$

2.2 Quotients

For more details on the contents of this section, we refer the reader to [ORSW22] and [ORSW25]. A $\mathbb{C}^*$-action on a polarized variety (X, L) allows one to define projective quotients of X, as follows. Set, for every $\tau \in [0, \delta] \cap \mathbb{Q}$,

$$A(\tau) := \bigoplus_{\substack{m \geq 0 \\ m\tau \in \mathbb{Z}}} \mathrm{H}^0(X, mL)_{m\tau}, \quad \mathcal{G}X(\tau) := \mathrm{Proj}(A(\tau)).$$

By [ORSW25, Section 2.2], $\mathcal{G}X(\tau)$ is the same for any $\tau \in (a_i, a_{i+1}) \cap \mathbb{Q}$, so we set

- $\mathcal{G}X_{i,i} := \mathcal{G}X(\tau)$ for $\tau = a_i$,
- $\mathcal{G}X_{i,i+1} := \mathcal{G}X(\tau)$ for $\tau \in (a_i, a_{i+1}) \cap \mathbb{Q}$.

Note that $\mathcal{G}X_{i,i+1}$ is a *geometric quotient* of X for every i, while $\mathcal{G}X_{i,i}$ is only a good categorical quotient (see [Dol03, Section 6.1, Definition, p. 94]). Following [BBCM02, Chapter 3, p. 10], we say that $\mathcal{G}X_{i,i}$ is *semigeometric*. Furthermore, the varieties $\mathcal{G}X_{i,i+1}$ are birationally equivalent, and the natural maps among them fit in the following commutative diagram, where the diagonal arrows are contractions and the horizontal maps are Atiyah flips (in the sense of [ORSW23b, Definition 6.1]):

$$\begin{array}{ccccccccc} & \mathcal{G}X_{0,1} & \dashrightarrow & \mathcal{G}X_{1,2} & \cdots & \mathcal{G}X_{r-2,r-1} & \dashrightarrow & \mathcal{G}X_{r-1,r} & \\ \swarrow & & \searrow \quad \swarrow & & & & \searrow \quad \swarrow & & \searrow \\ \mathcal{G}X_{0,0} & & \mathcal{G}X_{1,1} & \cdots & & \cdots & \mathcal{G}X_{r-1,r-1} & & \mathcal{G}X_{r,r}. \end{array} \tag{2.3}$$

2.2.1 Prunings

Given an equalized $\mathbb{C}^*$-action of criticality r and bandwidth δ on the smooth polarized pair (X, L), we consider the blowup $\beta\colon X^\flat \to X$ of X along the sink Y_0 and the source Y_r, with exceptional divisors $Y_0^\flat, Y_r^\flat$, and the induced $\mathbb{C}^*$-action. Given rational numbers $\tau_- < \tau_+$ in $[0, \delta]$, we set

$$L(\tau_-, \tau_+) := \beta^* L - \tau_- Y_0^\flat - (\delta - \tau_+) Y_r^\flat$$

and define the *pruning* of (X, L) with respect to τ_-, τ_+ as

$$X(\tau_-, \tau_+) := \mathrm{Proj} \bigoplus_{m \in k\mathbb{Z}_{\geq 0}} \mathrm{H}^0(X^\flat, mL(\tau_-, \tau_+)),$$

where $k > 0$ is the minimum integer such that $k\tau_\pm \in \mathbb{Z}_{>0}$. As shown in [ORSW25, Lemma 3.5], for values $\tau_-, \tau_+ \in ([0, \delta] \cap \mathbb{Q}) \setminus \{a_0, \dots a_r\}$, the pruning $X(\tau_-, \tau_+)$ depends only on the intervals (a_i, a_{i+1}), (a_{j-1}, a_j) containing τ_-, τ_+, respectively. Hence we can set

$$X_{i,j} := X(\tau_-, \tau_+) \quad \text{whenever } \tau_- \in (a_i, a_{i+1}),\ \tau_+ \in (a_{j-1}, a_j).$$

2.2.2 Chow quotient

Given a projective variety X with a nontrivial $\mathbb{C}^*$-action, the *normalized Chow quotient* of the action is the normalization $\mathcal{C}X$ of the closure of the subscheme of $\mathrm{Chow}(X)$ parametrizing the closures of general orbits.

More precisely, one may find a nonempty open set $V \subset X$ of points x whose orbits $\overline{\mathbb{C}^* \cdot x}$ belong to a single homology class in X. By shrinking V, if necessary, the geometric quotient of V by the action of $\mathbb{C}^*$ exists. Taking closures in X of the orbits of points of V, we obtain an algebraic irreducible family of rational curves in X and a morphism $\psi\colon V/\mathbb{C}^* \to \mathrm{Chow}(X)$.

The *Chow quotient* of X by the action of $\mathbb{C}^*$ is then defined as the closure

$$\overline{\mathcal{C}X} := \overline{\psi(V/\mathbb{C}^*)} \subset \mathrm{Chow}(X).$$

Its normalization $\mathcal{C}X$ will be called the *normalized Chow quotient* of X by the $\mathbb{C}^*$-action. Pulling back the universal family of $\mathrm{Chow}(X)$ to $\mathcal{C}X$, we obtain a family of 1-dimensional varieties $p\colon \mathcal{U} \to \mathcal{C}X$, with evaluation morphism $q\colon \mathcal{U} \to X$.

Remark 2.3. Let us consider a smooth polarized pair (X, L) endowed with an equalized $\mathbb{C}^*$-action of criticality r, with critical values $a_0, \dots, a_r$. By [ORSW25, Lemma 4.3], the fibers of p are reduced 1-dimensional schemes with smooth irreducible components, whose dual graphs are of type A. Thus any element of the family parametrized by $\mathcal{C}X$ takes the form $C = C_1 \cup \cdots \cup C_k$, where

(C1) every C_j is the (smooth) closure of a 1-dimensional orbit;

(C2) the singular points of C are the (transversal) intersections $P_j := C_j \cap C_{j+1}$ and are fixed points of weights a_{i_j}, with $0 < i_1 < \cdots < i_{k-1} < r$;

(C3) every P_j is the source of C_j and the sink of C_{j+1}, for $j = 1, \dots, k-1$;

(C4) the sink of C_1 and the source of C_k belong to Y_0 and Y_r, respectively.

Note that property (C4) follows from Lemma 2.1 together with the fact that the L-degree of C equals the L-degree of the general element of $\mathcal{C}X$, which is $\delta = a_r - a_0$.

Remark 2.4. From this description of the fibers of p, one may prove (see [ORSW25, Lemma 4.5]) that the family p is flat. In particular, $\mathcal{C}X$ is isomorphic to the normalization of an irreducible $\mathbb{C}^*$-invariant subvariety of $\mathrm{Hilb}(X)$, which we call the *Hilbert quotient* of the action (see [ORSW25, Remark 4.6]).

A priori, it is not clear that any $\mathbb{C}^*$-invariant subscheme $C = C_1 \cup \cdots \cup C_k$ satisfying the properties (C1)–(C4) listed in Remark 2.3 belongs to $\mathcal{C}X$. However, this has been proved to be true, under the following mild additional assumptions.

Setup 2.5. The pair (X, L) is a smooth polarized pair endowed with an equalized $\mathbb{C}^*$-action of criticality r, with critical values $0 = a_0 < \cdots < a_r$, such that $\nu^{\pm}(Y) \geq 2$ for every fixed-point component of weight a_i, with $2 \leq i \leq r-2$.

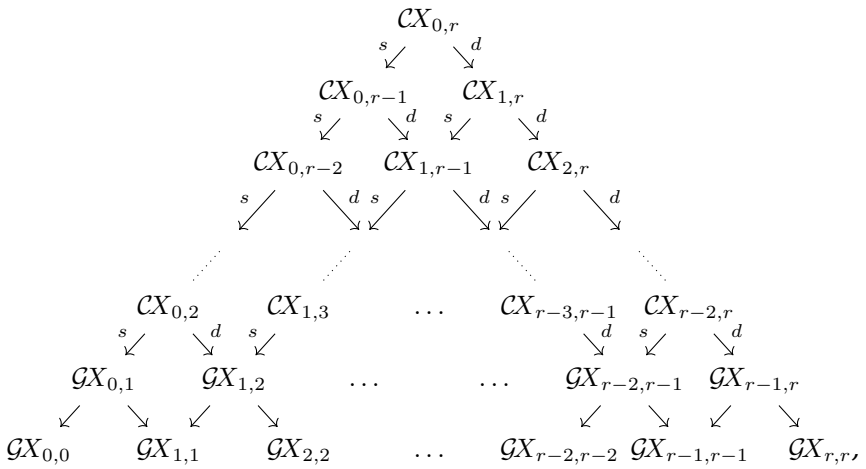
Remark 2.6. The technical assumption above on the $\nu^{\pm}(Y)$ is fulfilled if the variety X has Picard number 1 (*cf.* [ORSW23b, Lemma 2.8] and [ORSW22, Lemma 4.5]).

Corollary 2.7 (*cf.* [ORSW25, Corollary 5.9])**.** *Let (X, L) be as in Setup 2.5. Then every $\mathbb{C}^*$-invariant 1-dimensional reduced subscheme C of X with irreducible components $C_1, \dots, C_k$ satisfying the properties* (C1)–(C4) *listed in Remark 2.3, is an element of the family parametrized by $\mathcal{C}X$.*

The proof uses the recursive description of $\mathcal{C}X$ given in [ORSW25, Theorem 1.1 and Corollary 7.1], which we now recall.

Theorem 2.8. *Let (X, L) be as in Setup 2.5. Denote by $X_{i,j}$, $0 \leq i < j \leq r$, the corresponding prunings of X, and by $\mathcal{C}X_{i,j}$ their normalized Chow quotients. Then $\mathcal{C}X = \mathcal{C}X_{0,r}$, and we have a*

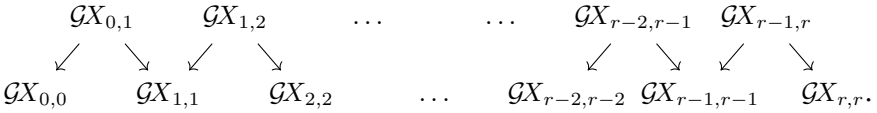
commutative diagram



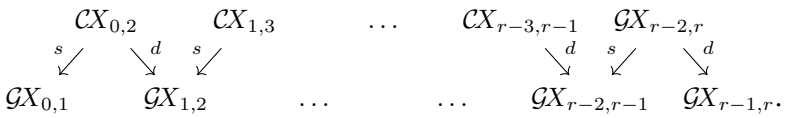
(2.4)

where all the arrows denoted by s, d are blowups and every rhombus in the diagram is a normalized Cartesian square.

In categorical language, we may simply say that $\mathcal{C}X$ is the normalized inverse limit of the GIT quotients



The centers of the blowups s, d appearing in Diagram (2.4) have been described in [ORSW25, Remark 5.10]; they are smooth for the maps appearing in the first layer right above the geometric quotients (see [ORSW25, Lemma 5.7]):



For the maps in successive layers, the centers are not necessarily smooth, as shown in [ORSW25, Example 6.1]). However, the situation improves under certain hypotheses. Let us recall the following.

Definition 2.9. A smooth projective variety X is said to be *convex* if for every morphism $\mu\colon \mathbb{P}^1 \to X$ we have that $\mathrm{H}^1(\mathbb{P}^1, \mu^* T_X) = 0$.

The following has been shown in [ORSW25, Theorem 1.2].

Theorem 2.10. *Let (X, L) be as in Setup 2.5. Assume moreover that X is convex. Then the varieties $\mathcal{C}X_{i,j}$ for $i = 0$ or $j = r$ are smooth, and the maps*

$$\mathcal{G}X_{0,1} \xleftarrow{s} \mathcal{C}X_{0,2} \xleftarrow{s} \cdots \xleftarrow{s} \mathcal{C}X \xrightarrow{d} \cdots \xrightarrow{d} \mathcal{C}X_{r-2,r} \xrightarrow{d} \mathcal{G}X_{r-1,r}$$

are blowups with smooth centers.

Let us illustrate Theorem 2.10 in the examples quoted in the introduction.

Example 2.11. The space of complete collineations from a vector space of dimension m to a vector space of dimension n, $n \leq m$, is the Chow quotient $\mathcal{C}X$ of the Grassmannian of $(n-1)$-dimensional linear subspaces of $\mathbb{P}^{n+m-1}$, which in the language of marked Dynkin diagrams we denote by $X = \mathrm{A}_{n+m-1}(n)$. We consider the action on this Grassmannian induced by the equalized action on $\mathbb{P}^{n+m-1}$ whose sink and source are $\mathbb{P}^{n-1}$ and $\mathbb{P}^{m-1}$. Following [ORSW23a, FSC24], one may easily obtain a description of all the fixed-point components of the action. In particular, one sees that the sink of the action on X is a point, so $\mathcal{G}X_{0,1}$ is the projective space $\mathbb{P}(\mathbb{C}^n \otimes \mathbb{C}^m)$. Then $\mathcal{C}X_{0,2}, \ldots, \mathcal{C}X_{0,n} = \mathcal{C}X$ are obtained by blowing up $\mathcal{G}X_{0,1}$ along the successive strict transforms of the traces of the closures of the BB-cells $X^+(Y_1), \ldots, X^+(Y_{n-1})$. These are precisely the successive strict transforms of the classes of matrices of ranks $1, \ldots, n-1$ in $\mathbb{P}(\mathbb{C}^n \otimes \mathbb{C}^m)$, so we recover the classical description of the space of collineations appearing in [Vai84].

Example 2.12. If we start with a quadric of equation

$$x_0x_n + \cdots + x_{n-1}x_{2n-1} = 0$$

in $\mathbb{P}^{2n-1}$, endowed with the equalized $\mathbb{C}^*$-action whose sink and source are the linear subspaces

$$Y_- : x_n = \cdots = x_{2n-1} = 0 \quad \text{and} \quad Y_+ : x_0 = \cdots = x_{n-1} = 0,$$

respectively, we may consider the induced action on the spinor variety $X = \mathrm{D}_n(n)$—more precisely, the one parametrizing the deformations of Y_-. The point corresponding to Y_- is the sink of the action on $\mathrm{D}_n(n)$, and the inner fixed-point components, as described in [FSC24], are Grassmannians, $\mathrm{A}_{n-1}(2s)$. The first geometric quotient $\mathcal{G}X_{0,1}$ is the projectivization of the tangent space of X at Y_-, which is $\mathbb{P}(\bigwedge^2 \mathbb{C}^n)$. As in the previous example, $\mathcal{C}X$ is then obtained by blowing up successively this projective space along the strict transforms of the traces of the closures of the BB-cells $X^+(Y_1), \ldots, X^+(Y_{\lfloor (n-1)/2 \rfloor})$, and one may check that these are precisely the strict transforms of the loci of classes of skew-forms $w \in \bigwedge^2 \mathbb{C}^n$ of ranks $2, \ldots, 2\lfloor (n-1)/2 \rfloor$. This is the standard construction of the space of complete skew-forms, as presented in [Tha99].

Example 2.13. Similarly, we may start with a Lagrangian Grassmannian $X = \mathrm{C}_n(n)$ parametrizing Lagrangian spaces $\mathbb{P}^{n-1}$ in a projective space $\mathbb{P}^{2n-1}$ with respect to a nondegenerate skew-symmetric form. We may assume that

$$Y_- : x_n = \cdots = x_{2n-1} = 0 \quad \text{and} \quad Y_+ : x_0 = \cdots = x_{n-1} = 0$$

are Lagrangian and consider the action on X induced by the equalized $\mathbb{C}^*$-action on $\mathbb{P}^{2n-1}$ whose sink and source are $Y_\pm$. The sink of the action in X is the point parametrizing Y_-, the geometric quotient $\mathcal{G}X_{0,1}$ is $\mathbb{P}(S^2\mathbb{C}^n)$, and the inner fixed components are Grassmannians $\mathrm{A}_{n-1}(1), \ldots, \mathrm{A}_{n-1}(n-1)$ (see [FSC24]). By a similar argument, we obtain that $\mathcal{C}X$ is the successive blowup of $\mathbb{P}(S^2\mathbb{C}^n)$ along the strict transforms of the classes of elements $q \in S^2\mathbb{C}^n$ of ranks $1, 2, \ldots, n-1$, which again coincides with the classical description given in [Vai82].

3 Chow quotients of convex varieties of Picard number 1

Throughout the rest of the paper, we will work in the following setting.

Setup 3.1. Let X be a smooth convex variety of Picard number 1, endowed with an equalized nontrivial $\mathbb{C}^*$-action of criticality r. We will denote by $\mathcal{M}$ a family of rational curves of minimal degree on X (which exists since X is uniruled), by $p\colon \mathcal{U} \to \mathcal{M}$ its universal family, and by $q\colon \mathcal{U} \to X$ the evaluation morphism (see [Kol96, Proposition II.2.11]).

Remark 3.2. Since X is convex, $\mathcal{M}$ is *beautiful*; *i.e.*, the evaluation morphism $q\colon \mathcal{U} \to X$ is smooth and surjective (see [OS26, Section 2.7] for further details). We will denote by $\mathcal{U}_x$ the fiber of q over the point $x \in X$. Note that since X is a Fano manifold (in particular, simply connected), then every $\mathcal{U}_x$ is connected, therefore irreducible (*cf.* [Kan17, Proposition 1.6]).

Let us recall some results recently obtained in [OS26]. The $\mathbb{C}^*$-action on X induces actions on $\mathcal{M}$ and $\mathcal{U}$ such that p and q are equivariant morphisms. These actions are equalized (see [OS26, Propositions 4.5 and 4.8]); in particular, the closure of every 1-dimensional orbit in $\mathcal{M}$ and $\mathcal{U}$ is a smooth rational curve.

Remark 3.3. Given a rational curve $\Gamma \subset \mathcal{M}$ that is the closure of a 1-dimensional orbit, $S_\Gamma := p^{-1}(\Gamma)$ is a rational ruled surface $\mathbb{F}_e$ with $e \leq 2$ (see [OS26, Proposition 4.6]). By the same proposition, if no point of Γ parametrizes a curve that is fixed by the action on X, then $e = 0$ or $e = 2$ and the action has four fixed points (see [OS26, Figure 4.1]).

The following statement has been proved in [OS26, Theorem 5.3].

Proposition 3.4. *In the situation of Setup 3.1, denoting by L the ample generator of $\mathrm{Pic}(X)$, the L-weights of the action are $0 < d < 2d < \cdots < (r-1)d < rd = \delta$, and, for every $j = 0, \ldots, r$, there exists a unique $Y_j \in \mathcal{Y}$ such that $\mu_L(Y_j) = jd$.*

Remark 3.5. Proposition 3.4 allows us to use the following notation:

$$B_j^\pm := B^\pm(Y_j), \;\; \nu_j^\pm := \nu^\pm(Y_j) \quad \text{for every } j = 0, \ldots, r \quad \text{and} \quad \nu_{-1}^+ = \nu_{r+1}^- := 0.$$

Note that, by definition, $\nu_0^+ = \nu_r^- = 0$, $\nu_0^- = \dim X - \dim Y_0$, and $\nu_r^+ = \dim X - \dim Y_r$.

3.1 Boundary divisors

The main result of this section is that the subscheme of $\mathcal{C}X$ parametrizing reducible cycles is a union of smooth divisors, which we call *boundary divisors*, with simple normal crossings.

Our first observation is the following.

Remark 3.6. Since X is a convex variety, $\mathcal{C}X$ is smooth by Theorem 2.10, which has been proven in [ORSW25, Proposition 6.4]. The arguments there, applied directly to $\mathcal{C}X$, show when $\mathcal{C}X$ is considered as a subscheme of $\mathrm{Hilb}(X)$ (see Remark 2.4), the convexity assumption implies that $\mathrm{Hilb}(X)$ is smooth at every point of $\mathcal{C}X$. Then $\mathcal{C}X$, which is the only $\mathbb{C}^*$-fixed-point component of $\mathrm{Hilb}(X)$ containing the class of the closure of the general orbit in X, is smooth as well.

Given a closed point $[C]$ of $\mathcal{C}X \subset \mathrm{Hilb}(X)$, it determines a $\mathbb{C}^*$-invariant subscheme C of X. Moreover, as observed in [ORSW25, Proof of Proposition 6.4], the singularities of C are nodes. Let us now recall some facts about deformations of projective nodal curves. We refer the interested reader to [Tal09, Section 6] for a detailed account on this topic.

Proposition 3.7. *Let C be a projective connected curve whose irreducible components are smooth rational curves $C_1, \dots, C_k$ and whose singular locus is the set of points $\{P_i = C_i \cap C_{i+1} |\ i = 1, \dots, k-1\}$. We assume that the singularities of C at these points are ordinary nodes. Then*

- *$\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)$ is supported on the singular points P_i, $i = 1, \dots, k-1$, and*

$$\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)_{P_i} \simeq \mathbb{C}$$

  for every $i = 1, \dots, k-1$;

- *the complex vector space $\mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$ of first-order deformations of C has dimension $k-1$, and it is naturally isomorphic to*

$$\mathrm{H}^0(C, \mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)) \simeq \bigoplus_{i=1}^{k-1} \mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)_{P_i}.$$

Proof. For the first assertion, we refer the reader to [Tal09, Remark 6.4.2]. The second claim follows from the local-to-global Ext spectral sequence. □

Remark 3.8. Proposition 3.7 can be interpreted geometrically as follows. The first-order deformations of C correspond to smoothings of C at its singular points (see [Tal09, Sections 6.3 and 6.4] for details). More precisely, for every i, the 1-dimensional vector space $\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)_{P_i}$ corresponds, in the space $\mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$ of first-order deformations of C, to deformations that smoothen the point P_i and are locally trivial around the P_j with $j \neq i$.

In this way, when we choose isomorphisms $\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)_{P_i} \simeq \mathbb{C}$ and denote by t_i, $i = 1, \dots, k-1$, the compositions $t_i \colon \mathrm{Ext}^1(\Omega_C, \mathcal{O}_C) \to \mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)_{P_i} \simeq \mathbb{C}$, the deformations of C that are locally trivial around P_i are given by the equation $t_i = 0$.

We will now go back to the case in which C is the subscheme of X determined by a closed point $[C] \in \mathcal{C}X$ and see that every first-order deformation of C extends to a deformation of C as a $\mathbb{C}^*$-invariant subscheme of X. More precisely, we will show the following.

Proposition 3.9. *Let $[C]$ be a closed point of $\mathcal{C}X$. Then the restriction to $T_{\mathcal{C}X,[C]}$ of the Kodaira–Spencer map*

$$T_{\mathrm{Hilb}(X),[C]} \longrightarrow \mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$$

is surjective.

Proof. As observed in [ORSW25, Proof of Proposition 6.4], the singularities of C are local complete intersections; then we have a short exact sequence

$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega_{X|C} \longrightarrow \Omega_C \longrightarrow 0 \tag{3.1}$$

and $\mathrm{Ext}^1(\Omega_{X|C}, \mathcal{O}_C) = 0$. We get the exact sequence

$$0 \longrightarrow \mathrm{Hom}(\Omega_C, \mathcal{O}_C) \longrightarrow \mathrm{Hom}(\Omega_{X|C}, \mathcal{O}_C) \longrightarrow \mathrm{Hom}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_C) \longrightarrow \mathrm{Ext}^1(\Omega_C, \mathcal{O}_C) \longrightarrow 0. \tag{3.2}$$

The space $\mathrm{Hom}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_C)$ is the Zariski tangent space of $\mathrm{Hilb}(X)$ at $[C]$, and the map $\mathrm{Hom}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_C) \to \mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$ is the Kodaira–Spencer map, associating with every first-order deformation of C as a

subscheme of X the corresponding first-order deformation of C as a scheme over $\mathrm{Spec}(\mathbb{C})$ (*cf.* [Ser06, Remark 2.4.4]). We then see that this map is surjective.

Now we add the action of $\mathbb{C}^*$ into the picture. On one hand, we have induced $\mathbb{C}^*$-actions on the spaces $\mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$ and $\mathrm{H}^0(C, \mathcal{E}xt^1(\Omega_C, \mathcal{O}_C))$, and the natural map between them is $\mathbb{C}^*$-equivariant. Hence, since $\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)$ is supported on the singular points of C, we conclude that the action on $\mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$ is trivial.

On the other hand, the tangent space $T_{\mathcal{C}X,[C]}$ is the weight zero eigenspace $\mathrm{Hom}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_C)_0 \subset \mathrm{Hom}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_C)$, which must surject onto the weight zero eigenspace $\mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)_0 = \mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$. This completes the proof. □

Corollary 3.10. *In the situation of Setup 3.1, the subscheme $\mathcal{C}X_{\mathrm{sing}}$ of $\mathcal{C}X$ parametrizing singular $\mathbb{C}^*$-invariant subschemes is a divisor with normal crossings.*

Proof. The fact that $\mathcal{C}X_{\mathrm{sing}}$ is a divisor follows from [Tal09, Propositions 6.2.11 and 6.3.5]. We will now study its local equations.

Let $[C]$ be a closed point, corresponding to a singular $\mathbb{C}^*$-invariant subscheme $C \subset X$, with irreducible components $C_1, \dots, C_k$ and singular points $P_i = C_i \cap C_{i+1}$, $i = 1, \dots, k-1$. Let us consider again the Zariski tangent space $T_{\mathcal{C}X,[C]}$ and the linear map

$$\phi \colon T_{\mathcal{C}X,[C]} \longrightarrow \mathrm{Ext}^1(\Omega_C, \mathcal{O}_C) \simeq \bigoplus_i \mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)_{P_i}$$

defined as the restriction to $T_{\mathcal{C}X,[C]}$ of the Kodaira–Spencer map, which is surjective by Proposition 3.9. We pull back via ϕ the linear coordinates $(t_1, \dots, t_{k-1})$ in $\mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$ introduced in Remark 3.8; abusing notation, we denote these linear forms by $t_i \colon T_{\mathcal{C}X,[C]} \to \mathbb{C}$. This tells us that locally around $[C]$ the divisor $\mathcal{C}X_{\mathrm{sing}} \subset \mathcal{C}X$ is given by an equation of the form $\bar{t}_1 \cdot \ldots \cdot \bar{t}_{k-1} = 0$, where $(d\bar{t}_i)_{[C]} = t_i$. In particular, it has normal crossings at $[C]$. □

To finish our description of $\mathcal{C}X_{\mathrm{sing}}$, we introduce the following divisors.

Definition 3.11. For every $i \in \{1, \dots, r-1\}$, we denote by $E_i^\circ \subset \mathcal{C}X$ the set parametrizing cycles having a unique singular point in the component Y_i. By Corollary 2.7 and the Białynicki-Birula theorem, E_i° is isomorphic to an open subset of $\mathbb{P}(N_{Y_i}^-) \times_{Y_i} \mathbb{P}(N_{Y_i}^+)$, which has dimension $n-1$. We denote by E_i its closure in $\mathcal{C}X$ and call it the ith *boundary divisor of* $\mathcal{C}X$.

Theorem 3.12. *Let X be a convex variety of Picard number 1, endowed with an equalized $\mathbb{C}^*$-action of criticality $r \geq 2$. Then the Chow quotient $\mathcal{C}X$ of X by the action of $\mathbb{C}^*$ is smooth, and the subscheme $\mathcal{C}X_{\mathrm{sing}} \subset \mathcal{C}X$ parametrizing reducible elements is a divisor with simple normal crossings, equal to*

$$\sum_{i=1}^{r-1} E_i.$$

Moreover, we have $[C] \in E_i$ if and only if the cycle C has a singular point in Y_i.

Remark 3.13. Note that in the case in which the criticality of the action is $r = 1$, Theorem 3.12 is obvious. In fact, in this case the Chow quotient $\mathcal{C}X$ equals the geometric quotient $\mathcal{G}X_{0,1}$, which is the projectivization of the normal bundle of Y_0 in X, hence smooth, and the subscheme of $\mathcal{C}X$ parametrizing reducible 1-cycles is empty.

Proof of Theorem 3.12. Take an element $[C] \in \mathcal{C}X$ consisting of k irreducible components $C_1, \dots, C_k$, and denote by $P_1 \in Y_{i_1}, \dots, P_{k-1} \in Y_{i_{k-1}}$ its singular points.

With the notation of the proof of Corollary 3.10, for every $j \in \{1, \dots, k-1\}$, there exists a smooth curve γ_j in $\mathcal{C}X$ passing through $[C]$, whose tangent space at $[C]$ is generated by $v \in T_{\mathcal{C}X,[C]}$, such that $\phi(v) \in \mathrm{Ext}^1(\Omega_C, \mathcal{O}_C)$ satisfies $t_j(\phi(v)) = 1$ and $t_s(\phi(v)) = 0$ for $s \neq j$.

The general element of γ_j has a unique singular point, which belongs to Y_{i_j}; in fact, the singular points of an element of $\mathcal{C}X$ are fixed points, and the unique singular point of an element of γ_j different from C is a deformation of $P_j \in Y_{i_j}$. In particular, the general element of γ_j belongs to E_{i_j}, so $[C] \in E_{i_j}$ as well. We conclude that $[C] \in \bigcap_{j=1}^{k-1} E_{i_j}$.

On the other hand, the arguments in the proof of Corollary 3.10 tell us that, locally around $[C]$, $\mathcal{C}X_{\mathrm{sing}}$ is the normal crossing intersection of exactly $k-1$ divisors; hence these divisors are necessarily the E_{i_j}. This finishes the proof. □

Remark 3.14. Theorem 3.12 is not true without the convexity assumption: In general, the subscheme of $\mathcal{C}X$ parametrizing reducible cycles may contain components different from the E_i defined above (see [ORSW25, Example 6.1]).

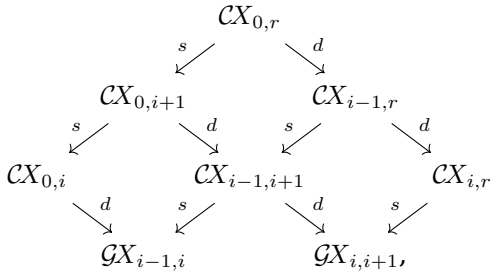
We now relate the boundary divisors with the exceptional divisors of the blowups appearing in Diagram (2.4). Note that the centers of the blowups $d\colon \mathcal{C}X_{i-1,i+1} \to \mathcal{G}X_{i,i+1}$ and $s\colon \mathcal{C}X_{i-1,i+1} \to \mathcal{G}X_{i-1,i}$ are smooth by [ORSW25, Lemma 5.7].

Lemma 3.15. *For every $i = 1, \dots, r-1$, let E_i' be the exceptional divisor of the blowups $d\colon \mathcal{C}X_{i-1,i+1} \to \mathcal{G}X_{i,i+1}$, $s\colon \mathcal{C}X_{i-1,i+1} \to \mathcal{G}X_{i-1,i}$. Then the pullback of E_i' to $\mathcal{C}X$ is the boundary divisor E_i.*

Proof. Let $[C] \in \mathcal{C}X$ be a cycle; its image in $\mathcal{C}X_{i-1,i+1}$ can be described as follows: We take the intersection of C with the open set $X \setminus (B_{i-1}^+ \cup B_{i+1}^-)$, which maps isomorphically into the pruning $X_{i-1,i+1}$, and we consider its closure in $X_{i-1,i+1}$; this determines an element $[C'] \in \mathcal{C}X_{i-1,i+1}$. The divisor E_i' is isomorphic to $\mathbb{P}(N_{Y_i}^-) \times_{Y_i} \mathbb{P}(N_{Y_i}^+)$; hence $[C'] \in E_i'$ if and only if C is singular at a point of Y_i, By Theorem 3.12, this happens if and only if $[C] \in E_i$.

This shows that the inverse image of E_i' (which is the support of the pullback of E_i' to $\mathcal{C}X$) is E_i. It also follows that the birational map $\mathcal{C}X \to \mathcal{C}X_{i-1,i+1}$ is a local isomorphism at a general point of E_i; hence E_i is the pullback of E_i'. □

As a consequence of Lemma 3.15, considering the commutative diagram with Cartesian squares



we see that E_i equals both the pullbacks to $\mathcal{C}X = \mathcal{C}X_{0,r}$ of the exceptional divisors of the smooth blowups $s\colon \mathcal{C}X_{0,i+1} \to \mathcal{C}X_{0,i}$ and $d\colon \mathcal{C}X_{i-1,r} \to \mathcal{C}X_{i,r}$.

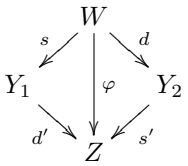
3.2 Smoothness of the partial Chow quotients

Here we will show that, in the situation of Setup 3.1, all the maps s, d appearing in Diagram (2.4) are blowups with smooth centers.

Theorem 3.16. *Let X be as in Setup 3.1. Then, for $i < j$, the Chow quotients $\mathcal{C}X_{i,j}$ are smooth, and the maps $s\colon \mathcal{C}X_{i,j+1} \to \mathcal{C}X_{i,j}$ and $d\colon \mathcal{C}X_{i-1,j} \to \mathcal{C}X_{i,j}$ are blowups with smooth irreducible centers.*

The proof of Theorem 3.16 will be obtained by using recursively the following.

Lemma 3.17. *Consider a Cartesian diagram of normal projective varieties*



such that

- *W, Y_1, Y_2 are smooth, and the morphisms s, d are blowups with smooth irreducible centers and exceptional loci E_s, E_d;*
- *$\varphi(E_s) \cap \varphi(E_d)$ is a proper subset of $\varphi(E_s)$ and of $\varphi(E_d)$.*

Then Z is smooth, and the maps d', s' are blowups with smooth centers.

Proof. It is clearly enough to show that d' is a blowup with smooth center. By our second assumption, $s(E_d)$ is not contained in $s(E_s)$; hence we may assert that d' is a divisorial contraction with exceptional locus $s(E_d)$. This is elementary because the relative Picard number $\rho(W/Z)$ is 2, since φ contracts a 2-dimensional face of the Mori cone of W.

Moreover, there exists a point $P \in d'(\mathrm{Exc}(d')) \setminus d'(s(E_s))$ such that s is a local isomorphism at the points of the fiber $F = \varphi^{-1}(P)$. Since the diagram is Cartesian and d is a smooth blowup, then F is a projective space. Therefore,

$$(-K_{Y_1})_{|s(F)} = (-s^* K_{Y_1})_{|F} = (-K_W)_{|F} = \dim F = \dim s(F).$$

Then, by [AO02, Theorem 5.1], it follows that d' is a blowup with a smooth irreducible center. □

Proof of Theorem 3.16. From Theorem 2.10, we already know that the assertion is true for $i = 0$ and $j = r$, that is, for the varieties and maps appearing on the upper sides of Diagram (2.4). Now we apply Lemma 3.17 recursively on $r - i + j$.

At a given step, if $s\colon \mathcal{C}X_{i-1,j+1} \to \mathcal{C}X_{i-1,j}$ and $d\colon \mathcal{C}X_{i-1,j+1} \to \mathcal{C}X_{i,j+1}$ are blowups with smooth centers, we consider the diagram

$$\begin{array}{ccccc} & & \mathcal{C}X_{i-1,j+1} & & \\ & \overset{s}{\swarrow} & & \overset{d}{\searrow} & \\ \mathcal{C}X_{i-1,j} & & & & \mathcal{C}X_{i,j+1} \\ & \searrow & & \swarrow & \\ & & \mathcal{C}X_{i,j}. & & \end{array}$$

The exceptional divisors of s, d are the images in $\mathcal{C}X_{i-1,j+1}$ of $E_j, E_i \subset \mathcal{C}X$, respectively. In particular, they are irreducible. By Theorem 3.12, the intersection of these two sets is properly contained in both. This concludes the proof. □

3.3 Primal curves

In this section, we construct some special curves in $\mathcal{C}X$; we will show later that their classes generate the Mori cone of $\mathcal{C}X$. The construction is based on the following.

Lemma 3.18. *For every $i = 0, \dots, r-1$ and for every point $P_{i-1} \in Y_{i-1}$, there exist a point $P_{i+1} \in Y_{i+1}$ and a curve $\Gamma \subset \mathcal{M}$, which is the closure on an orbit, such that $S = p^{-1}(\Gamma)$ is a ruled surface $\mathbb{F}_2$, the sink and the source of the action on S are mapped by the evaluation map $q\colon \mathcal{U} \to X$ to P_{i-1} and to P_{i+1}, respectively, and the minimal section is contracted by q to a point $P_i \in Y_i$.*

Proof. Take a general $\mathbb{C}^*$-invariant curve C_- in $\mathcal{M}$ connecting $P_{i-1} \in Y_{i-1}$ to Y_i, and denote its source by $P_i \in Y_i$. The existence of C_- is guaranteed by [OS26, Theorem 5.3]. Now consider the action of $\mathbb{C}^*$ on the invariant subvariety $p(\mathcal{U}_{P_i}) \subset \mathcal{M}$. By construction, the element $[C_-]$ belongs to the sink of the action on $p(\mathcal{U}_{P_i})$. Let $\Gamma \subset p(\mathcal{U}_{P_i})$ be the closure of a general orbit with sink $[C_-]$. Its source $[C_+]$ parametrizes a $\mathbb{C}^*$-invariant curve C_- in $\mathcal{M}$ with sink P_i and source $P_{i+1} \in Y_{i+1}$. In particular, we have constructed a $\mathbb{C}^*$-invariant connected cycle $C_{\mathrm{s}} = C_- + C_+$ linking $P_{i-1} \in Y_{i-1}$ to $P_{i+1} \in Y_{i+1}$.

Consider $S = p^{-1}(\Gamma)$; since the induced action on $\mathcal{M}$ is equalized (see [OS26, Proposition 4.5]), S is a smooth ruled surface. Since $C_\pm$ are not fixed and $P_i \in C$, by [OS26, Proposition 4.6], we have $S \simeq \mathbb{F}_2$. The action is the one described in Remark 3.3. □

Let us set

$$I := \{k \mid \dim Y_k > 0\} \subseteq \{0, \dots, r\}. \tag{3.3}$$

Construction 3.19. Given $i \neq 0, r$, we consider a point $P_{i-1} \in Y_{i-1}$ and the 1-dimensional family of orbits linking P_{i-1} to a point P_{i+1} in Y_{i+1} constructed in Lemma 3.18. Closing these orbits and adding $\mathbb{C}^*$-invariant cycles Δ_0 linking P_{i-1} to the sink and Δ_r linking P_{i+1} to the source, we obtain a curve $\Gamma_i \subset \mathcal{C}X$.

In fact, the Chow quotient of the ruled surface $\mathbb{F}_2$ with respect to the $\mathbb{C}^*$-action described in Remark 3.3 is isomorphic to the projectivization of the tangent space at its sink, *i.e.*, to $\mathbb{P}^1$, and the equivariant morphism $\mathbb{F}_2 \to X$ given by Lemma 3.18 induces a morphism $\mathbb{P}^1 \to \mathcal{C}X_{i-1,i+1}$. Moreover, we have two constant maps $\mathbb{P}^1 \to \mathcal{C}X_{0,i-1}$ and $\mathbb{P}^1 \to \mathcal{C}X_{i+1,r}$, defined by the choice of Δ_0 and Δ_r. Since $\mathcal{C}X$ is the normalized inverse limit of Diagram (2.4), we get a morphism from $\mathbb{P}^1$ to $\mathcal{C}X$.

In a similar way, if 0 or r belongs to I, we can construct rational curves $\Gamma_0, \Gamma_r \subset \mathcal{C}X$. Let us explain the case $i = 0$ (the case $i = r$ is analogous).

If $0 \in I$, we consider a point $P_1 \in Y_1$ and a line $\ell \subset \mathbb{P}(\mathcal{N}(Y_1)^+_{P_1})$. For every element of ℓ, we have a $\mathbb{C}^*$-invariant curve in $\mathcal{M}$ linking P_1 to Y_0. Adding a fixed $\mathbb{C}^*$-invariant cycle Δ_r linking P_1 and the source, we obtain a 1-dimensional family of $\mathbb{C}^*$-invariant cycles, parametrized by a rational curve $\Gamma_0 \subset \mathcal{C}X$.

Definition 3.20. We will call *primal curves* in $\mathcal{C}X$ the curves $\Gamma_i, i \in I$, described in Construction 3.19. We have represented the families of cycles parametrized by these curves in Figure 1.

3.4 Intersection numbers

In this section, we compute the intersection matrix of primal curves and boundary divisors.

We first note that the cycles parametrized by Γ_0 (respectively, Γ_r) are singular only at points of Y_1 (respectively, Y_{r-1}); hence $E_i \cdot \Gamma_0 = 0$ for $i \geq 1$ (respectively, $E_i \cdot \Gamma_r = 0$ for $i \leq r$). On the other hand, the cycles parametrized by Γ_j can only be singular at points of Y_{j-1}, Y_j, Y_{j+1}; therefore, $E_i \cdot \Gamma_j = 0$ for $i \neq j-1, j, j+1$. We are left with computing the values a_{ij} for $i = j$, $i = j-1$, and $i = j+1$.

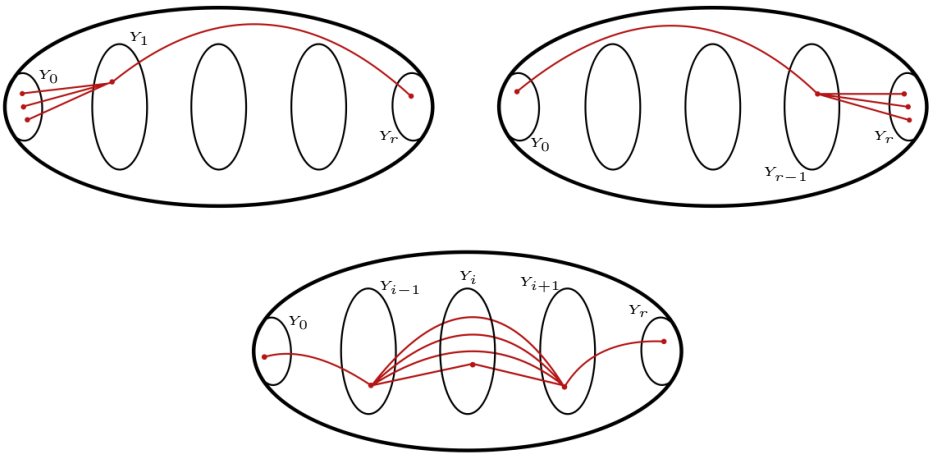


Figure 1: The 1-dimensional families of $\mathbb{C}^*$-invariant cycles determining the curves $\Gamma_0, \Gamma_r, \Gamma_i$ ($i = 1, \dots, r-1$).

Lemma 3.21. *We have $E_i \cdot \Gamma_j = -1$ if $j = i-1, i+1$.*

Proof. We will show the statement in the case $j = i+1$; the case $j = i-1$ is analogous. We consider the following diagram, with Cartesian squares:

$$\begin{array}{ccccc} & & \mathcal{C}X_{0,r} & & \\ & \swarrow^{s} & & \searrow^{d} & \\ \mathcal{C}X_{0,i+1} & & & & \mathcal{C}X_{i-1,r} \\ & \searrow_{d} & & \swarrow_{s} \quad \searrow^{d} & \\ & & \mathcal{C}X_{i-1,i+1} & & \mathcal{C}X_{i,r} \\ & & & \searrow^{d} \quad \swarrow_{s} & \\ & & & \mathcal{G}X_{i,i+1}. & \end{array}$$

We claim that the image Γ'_{i+1} of Γ_{i+1} in $\mathcal{C}X_{i-1,i+1}$, via the isomorphism $E'_i \simeq \mathbb{P}(N^+(Y_i)) \times_{Y_i} \mathbb{P}(N^-(Y_i))$ given by the Białynicki-Birula theorem, is a line in a fiber of the projection $\mathbb{P}(N^+(Y_i)) \times_{Y_i} \mathbb{P}(N^-(Y_i)) \to \mathbb{P}(N^+(Y_i))$.

In fact, the cycles of the family Γ_{i+1} are constructed by means of an equivariant morphism $\mathbb{F}_2 \to X$, so the sink x_- of the action on $\mathbb{F}_2$ maps to a point $P_i \in Y_i$ contained in all the cycles parametrized by Γ_{i+1}. Since the action of $\mathbb{C}^*$ on X is equalized, the cycles parametrized by Γ_{i+1} must have different tangent directions at P_i, and we may then conclude that the differential of the map $\mathbb{F}_2 \to X$ at x_- is injective. The line obtained as the projectivization of its image, which lies in $\mathbb{P}(N^-(Y_i)_{P_i})$, is the curve Γ'_{i+1}.

By Lemma 3.15, E_i is the pullback to $\mathcal{C}X$ of the exceptional divisor E'_i of the map $d\colon \mathcal{C}X_{i-1,i+1} \to \mathcal{G}X_{i,i+1}$. On the other hand, Γ'_{i+1} is a line in a fiber of the contraction of $E'_i \simeq \mathbb{P}(N^-(Y_i)) \times_{Y_i} \mathbb{P}(N^+(Y_i)) \to \mathbb{P}(N^+(Y_i))$. Since the map d is a blowup with a smooth center, we conclude that

$$E_i \cdot \Gamma_{i+1} = E'_i \cdot \Gamma'_{i+1} = -1. \qquad \square$$

Lemma 3.22. *We have $E_i \cdot \Gamma_i = 2$ for every $i = 1, \dots, r-1$.*

Proof. Consider a curve Γ_i in $\mathcal{C}X$. There exists precisely one element C_s in this family of cycles meeting the fixed-point component Y_i (see Lemma 3.18). Let us denote the intersection point by P_i and note that two components of C_s are $\mathbb{C}^*$-invariant curves of $\mathcal{M}$ linking P_i to points $P_{i-1} \in Y_{i-1}$ and $P_{i+1} \in Y_{i+1}$, respectively. We now use [BB73, Theorem 2.5] to claim that X is locally $\mathbb{C}^*$-equivariantly isomorphic to $U \times N_{Y_i|X,P_i}$ for an open neighborhood U of P_i in Y_i.

We consider linear coordinates $(x_1, \dots, x_{\nu_i^-})$ and $(y_1, \dots, y_{\nu_i^+})$ in the normal spaces $N(Y_i)^-_{P_i}$, $N(Y_i)^+_{P_i}$, respectively, and linear coordinates $z_1, \dots, z_k$ of Y_i around P_i. We can appropriately choose these sets of coordinates so that, on a neighborhood of P_i, the cycles parametrized by the curve Γ_i have equations

$$x_1 y_1 + z_1^2 = z_1 + c = 0 \quad \text{and} \quad x_i = y_i = z_i = 0 \quad (i \geq 2),$$

where c is a complex parameter. This family (whose parameter space is an open set of Γ_i) contains a unique reducible conic (given by $c = 0$). We will compute the multiplicity of intersection of Γ_i with the set of reducible conics in the 3-dimensional space of coordinates x_1, y_1, z_1 at $c = 0$.

In order to do so, we first note that the linear span of a deformation of a conic of equations $x_1y_1 + z_1^2 = z_1 + c = 0$ has equation of the form $z_1 + c = ax_1 + by_1$ with $a, b \in \mathbb{C}$. In each of these planes, we have a distinguished conic with equations $x_1y_1 + z_1^2 = z_1 + c - ax_1 - by_1 = 0$. By adding the deformations of these conics within their linear span, we get a parametrization of the deformations of our conic $(x_1y_1 = z_1^2 = 0)$ in the 3-dimensional space:

$$x_1y_1 + z_1^2 + \big(dx_1^2 + ey_1^2 + gx_1 + hy_1 + f\big) = z_1 + c - ax_1 - by_1 = 0, \quad a, b, c, d, e, f, g, h \in \mathbb{C},$$

which can be written as

$$\begin{pmatrix} x_1 & y_1 & 1 \end{pmatrix} \begin{pmatrix} d + a^2 & 1 + ab & g - ac \\ 1 + ab & e + b^2 & h - bc \\ g - ac & h - bc & f + c^2 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \\ 1 \end{pmatrix} = z_1 + c - ax_1 - by_1 = 0.$$

Such a conic is reducible if and only if

$$-f - c^2 + (\text{terms of degree} \geq 3) = 0.$$

Restricting this equation to Γ_i (which is given by $a = b = d = e = f = g = h = 0$), we get $c^2 = 0$. □

Summing up, we may write the intersection numbers of the divisor E_i and the curves Γ_j in the matrix in the following statement (note that Γ_0 and Γ_r may not be defined if the sink and the source are isolated points).

Proposition 3.23. *Let X be as in Setup 3.1. Then the intersection matrix of the boundary divisors and the primal curves is the following:*

$$[E_i \cdot \Gamma_j] = \begin{pmatrix} -1 & 2 & -1 & 0 & \dots & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & \dots & 0 & 0 & 0 \\ 0 & 0 & -1 & \ddots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & \dots & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & \dots & 0 & -1 & 2 & -1 \end{pmatrix}$$

with $i = 1, \dots, r-1$ and $j = 0, \dots, r$.

4 Nef and Mori cones of the Chow quotient

We want now to describe the nef cone and the cone of curves of $\mathcal{C}X$, for a variety X as in Setup 3.1. Let us start by introducing some line bundles on $\mathcal{C}X$:

Notation 4.1. For every $k \in \{0, \dots, r\}$, let $\mathcal{L}_{k,k} \in \mathrm{Pic}(\mathcal{C}X)$ be the pullback of an ample line bundle $\mathcal{H}_{k,k}$ on $\mathcal{G}X_{k,k}$. Note that $\mathcal{L}_{0,0}$ and $\mathcal{L}_{r,r}$ may be trivial (if Y_0 or Y_r, respectively, are isolated points), and that the set $I = \{k \mid \dim Y_k > 0\} \subseteq \{0, \dots, r\}$ introduced in (3.3) can be written as $I = \{k \mid \mathcal{L}_{k,k} \not\simeq \mathcal{O}_{\mathcal{C}X}\}$.

We set

$$\mathcal{A} := \bigotimes_k \mathcal{L}_{k,k} \quad \text{and} \quad \mathcal{D}_i := \bigotimes_{k \neq i} \mathcal{L}_{k,k} = \mathcal{A} \otimes \mathcal{L}_{i,i}^{-1} \quad \text{for every } i \in \{0, \dots, r\}.$$

As in Definition 3.11, we will denote by $E_i \subset \mathcal{C}X$ the closure of the set parametrizing cycles having a unique singular point in Y_i, $i \in \{1, \dots, r-1\}$; note that, by Theorem 3.12, E_i parametrizes *all* the cycles having a singular point in Y_i, and $\sum E_i$ is a simple normal crossing divisor, whose support is the subset $\mathcal{C}X_{\mathrm{sing}} \subset \mathcal{C}X$ parametrizing singular $\mathbb{C}^*$-invariant cycles. In particular, every E_i is smooth. Abusing notation, we will denote by $E_i \in \mathrm{Pic}(\mathcal{C}X)$ the associated line bundles as well. Moreover, we will denote by $\mathcal{L}_{k,k}, \mathcal{A}, \mathcal{D}_i, E_i \in \mathrm{N}^1(\mathcal{C}X)$ the corresponding numerical classes of these divisors.

We start by determining the Picard number of $\mathcal{C}X$ and a basis of $\mathrm{N}^1(\mathcal{C}X)$.

Lemma 4.2. *The Picard number of $\mathcal{C}X$ is $|I|$, and $\mathrm{N}^1(\mathcal{C}X)$ is generated by the classes $\mathcal{L}_{i,i}$, $i \in I$.*

Proof. Let us first assume that $\dim Y_0, \dim Y_r > 0$. By [ORSW25, Theorem 1.2 and Remark 5.10], $\mathcal{C}X$ is obtained from the geometric quotient $\mathcal{G}X_{0,1}$ via a sequence of blowups

$$\mathcal{G}X_{0,1} \xleftarrow{s} \mathcal{C}X_{0,2} \xleftarrow{s} \quad \cdots \quad \xleftarrow{s} \mathcal{C}X_{0,r} = \mathcal{C}X$$

with smooth centers $S_{0,j} \subset \mathcal{C}X_{0,j}$, $j = 1, \dots, r-1$. Since there is only one fixed-point component of the action on X for each weight, the centers $S_{0,j}$ are irreducible; hence, for each $j = 1, \dots, r-1$, the contraction $s\colon \mathcal{C}X_{0,j+1} \to \mathcal{C}X_{0,j}$ is elementary. Then the statement about the Picard number follows by observing that $\rho(\mathcal{C}X_{0,1}) = 2$; in fact, $\mathcal{C}X_{0,1} = \mathcal{G}X_{0,1} = \mathbb{P}_{Y_0}(\mathcal{N}_{Y_0,X})$ and $\rho(Y_0) = 1$ by [ORSW23b, Lemma 2.8].

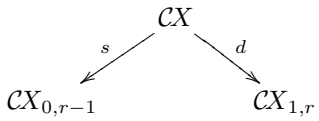
By abuse of notation, we will denote by $\mathcal{L}_{i,i}$ the class of the pullback of $\mathcal{H}_{i,i}$ to any Chow quotient $\mathcal{C}X_{0,j}$, $j \geq i$. To show the linear independence of $\{\mathcal{L}_{i,i}\}_{i\in I} \subset \mathrm{N}^1(\mathcal{C}X)$, we proceed by induction. First we observe that $\mathcal{L}_{0,0}$ and $\mathcal{L}_{1,1}$ are independent in $\mathrm{N}^1(\mathcal{C}X_{0,1})$ since they support two different contractions of $\mathcal{C}X_{0,1}$. Now assume that $\{\mathcal{L}_{i,i}\}_{i\leq k-1}$ is linearly independent in $\mathrm{N}^1(\mathcal{C}X_{0,k-1})$; since the map $\mathrm{N}^1(\mathcal{C}X_{0,k-1}) \to \mathrm{N}^1(\mathcal{C}X_{0,k})$ is injective, $\{\mathcal{L}_{i,i}\}_{i\leq k-1}$ is linearly independent in $\mathrm{N}^1(\mathcal{C}X_{0,k})$. To prove that $\{\mathcal{L}_{i,i}\}_{i\leq k}$ is linearly independent in $\mathrm{N}^1(\mathcal{C}X_{0,k})$, it is then enough to notice that, for every $i \leq k-1$, $\mathcal{L}_{i,i}$ is trivial on the fibers of $\mathcal{C}X_{0,k} \to \mathcal{C}X_{0,k-1}$, while $\mathcal{L}_{k,k}$ is ample on them. In fact, $\mathcal{C}X_{0,k}$ is the normalization of $\mathcal{C}X_{0,k-1} \times_{\mathcal{C}X_{k-1,k-1}} \mathcal{C}X_{k-1,k}$, so the fibers of $\mathcal{C}X_{0,k} \to \mathcal{C}X_{0,k-1}$ are, up to a finite morphism, inverse images of fibers of $\mathcal{C}X_{k-1,k} \to \mathcal{C}X_{k-1,k-1}$, which is an elementary contraction different from $\mathcal{C}X_{k-1,k} \to \mathcal{C}X_{k,k}$.

If $\dim Y_0 = 0$ and $\dim Y_r > 0$ (resp. $\dim Y_0 > 0$ and $\dim Y_r = 0$, resp. $\dim Y_0 = 0$ and $\dim Y_r = 0$), the statement follows applying the arguments above to the pruning $X_{1,r}$ (resp. $X_{0,r-1}$, resp. $X_{1,r-1}$). □

The line bundle $\mathcal{A} \in \mathrm{Pic}(\mathcal{C}X)$ is ample by [ORSW25, Lemma 6.4]. Furthermore, applying the same lemma to the prunings $X_{i,j}$ of X, we get the following.

Lemma 4.3. *For every $i < j$, $i, j \in \{0, \dots, r\}$, the natural morphism $\mathcal{C}X \to \mathcal{C}X_{i,j}$ is supported by the nef line bundle $\bigotimes_{i\leq k\leq j} \mathcal{L}_{k,k}$.*

Remark 4.4. In particular, the maps s, d in



are supported by the nef divisors $\mathcal{D}_r$, $\mathcal{D}_0$ and contract the curves Γ_r, Γ_0, respectively. For this reason, we will denote these maps by π_r and π_0, respectively. Note that the above statement together with [ORSW23b, Lemma 2.8] imply that the following conditions are equivalent:

- $\dim Y_0 = 0$.
- $\mathcal{L}_{0,0}$ is trivial.
- $\pi_0\colon \mathcal{C}X \to \mathcal{C}X_{1,r}$ is an isomorphism.
- $\nu_1^- = 1$.

In the case in which $\dim Y_0 > 0$, the exceptional locus of π_0 is E_1, and the fibers of $(\pi_0)_{|E_1}\colon E_1 \to \pi_0(E_1)$ are projective spaces of dimension $\nu_1^+ - 1$ (see the proof of [ORSW25, Lemma 5.8]). An analogous equivalence holds for $\pi_r\colon \mathcal{C}X \to \mathcal{C}X_{0,r-1}$.

In particular, if $\dim Y_0 > 0$ (resp. if $\dim Y_r > 0$), $\mathrm{Nef}(\mathcal{C}X)$ has a facet corresponding to the elementary contraction supported by $\mathcal{D}_r$ (resp. $\mathcal{D}_0$). We will now show that the remaining facets of $\mathrm{Nef}(\mathcal{C}X)$ are determined by the nef divisors $\mathcal{D}_i$, $i \in \{1, \dots, r-1\}$, introduced above. Note that $\mathcal{D}_i$ is the supporting divisor of the induced morphism

$$\pi_i\colon \mathcal{C}X \longrightarrow \prod_{j\neq i} \mathcal{G}X_{j,j}.$$

Proposition 4.5. *For every $i \in \{1, \dots, r-1\}$, the line bundle $\mathcal{D}_i$ is nef and not ample. Moreover, the associated morphism $\pi_i\colon \mathcal{C}X \to \prod_{j\neq i} \mathcal{G}X_{j,j}$ contracts the primal curves of type Γ_i to a point, and its exceptional locus is contained in $E_{i-1} \cap E_{i+1}$ (setting $E_0 = E_r = \mathcal{C}X$).*

Proof. We will show that for every i, the induced map $\pi_i\colon \mathcal{C}X \to \prod_{j\neq i} \mathcal{G}X_{j,j}$ is not finite. Two cycles $[C], [C'] \in \mathcal{C}X$ have the same image via π_i if and only if their projections to $\mathcal{C}X_{0,i-1}$ and $\mathcal{C}X_{i+1,r}$ are the same, that is, if the following three conditions hold:

- Their fixed points of weights different from i coincide.
- Their irreducible components with sink in Y_j, $j < i-1$, are equal.

- Their irreducible components with source in Y_j, $j > i+1$, are equal.

In particular, if $[C] \neq [C']$ have the same image via π_i, then they are singular at points of Y_{i-1} and Y_{i+1}, have the same fixed points in those components (say P_{i-1}, P_{i+1}), their subcycles contained in B^+_{i-1} and B^-_{i+1} are equal, while their subcycles joining P_{i-1} and P_{i+1} are different. This, on one hand, tells us that the exceptional locus of π_i is contained in $E_{i-1} \cap E_{i+1}$, and on the other that π_i contracts the primal curve Γ_i. This concludes the proof. □

Remark 4.6. By Proposition 4.5, the contractions π_i are small if $2 \leq i \leq r-2$. Moreover, we already know that π_i is a smooth blowup if i is the minimum or maximum index belonging to I. On the other hand, in the case in which $0 \in I$ (resp. $r \in I$), the contraction π_1 (resp. π_{r-1}) can be small, divisorial, or of fiber type, as shown in the following examples (see also Remark 4.11 below).

Example 4.7 (Fiber type). Consider the projective space $\mathrm{A}_5(1) = \mathbb{P}^5$, polarized with the hyperplane line bundle, endowed with the equalized bandwidth 1 action whose extremal fixed-point components are two disjoint linear subspaces Λ_- and Λ_+ of dimension 2. We consider the induced action on

$$X := \mathrm{A}_5(2) = \mathbb{G}(1,5).$$

The action on $\mathrm{A}_5(2)$ has bandwidth 2 with respect to the Plücker line bundle, and three fixed-point components; the extremal ones, Y_0 and Y_2, are the subvarieties parametrizing lines contained in Λ_- or Λ_+, both isomorphic to $\mathbb{P}^2$. The Chow quotient of X has dimension 7; hence the morphism $\pi_1 \colon \mathcal{C}X \to Y_0 \times Y_2$ is necessarily of fiber type.

Example 4.8 (Divisorial). Let us consider the odd-dimensional quadric $\mathrm{B}_5(1) = \mathbb{Q}^9 \subset \mathbb{P}^{10}$, polarized with the restriction of hyperplane line bundle, endowed with the equalized action that has isolated sink and source (see [FSC24, Section 3.2]) and the induced action of the corresponding (14-dimensional) orthogonal Grassmannian of lines in the quadric

$$X = \mathrm{B}_5(2).$$

The action on $\mathrm{B}_5(1)$ has isolated extremal fixed points z_-, z_+ and a unique inner fixed-point component Z_0, isomorphic to a smooth quadric $\mathrm{B}_4(1)$, which is the intersection of $\mathrm{B}_5(1)$ with the projective tangent spaces at z_- and z_+. The action on $\mathrm{B}_5(2)$ has three fixed-point components: The extremal ones are the subvarieties parametrizing lines of $\mathrm{B}_5(1)$ passing through z_- or z_+; they are both isomorphic to $\mathrm{B}_4(1) = \mathbb{Q}^7$. The inner one is the subvariety parametrizing lines contained in Z_0, which is isomorphic to $\mathrm{B}_4(2)$.

Given two points $y_0 \in Y_0$, $y_2 \in Y_2$ corresponding to two disjoint lines ℓ_0, ℓ_2, the linear span of ℓ_0 and ℓ_2 cuts $\mathrm{B}_5(1)$ in a smooth quadric surface. The conic in $\mathrm{B}_5(2)$ parametrizing the lines in the ruling containing ℓ_0 and ℓ_2 is the closure of an orbit with sink y_0 and source y_2, and it defines an element of $\mathcal{C}X$, which is the unique cycle passing through y_0 and y_2. In particular, this shows that the morphism $\mathcal{C}X \to Y_0 \times Y_2$, which assigns to every invariant cycle its intersection with Y_0 and Y_2 is surjective, hence, computing dimensions, birational.

On the other hand, given $y_0 \in Y_0$, there exist a unique point $y_0^* \in Y_2$ such that the lines ℓ_0, ℓ_0^* parametrized by these two points meet, at a point $z_0 \in Z_0$. Given a general 3-dimensional linear space Λ containing ℓ_0 and ℓ_0^* and contained in the projective tangent space $\mathbb{T}_{z_0}\mathbb{Q}^9$, its intersection with $\mathbb{Q}^9$ is a quadric cone containing ℓ_0 and ℓ_0^*. The lines in this cone are parametrized by a smooth conic in $\mathrm{B}_5(2)$, which is the closure of an orbit with sink y_0 and source y_0^*. An easy computation shows that there is a 6-dimensional family of such curves.

Hence the morphism $\pi_1 \colon \mathcal{C}X \to Y_0 \times Y_2$ has positive-dimensional fibers over the subvariety (y_0, y_0^*), which has dimension $7 = \dim Y_0 = \dim Y_2$, and these fibers have dimension 6. We conclude that π_1 has exceptional locus of dimension $13 = \dim \mathcal{C}X - 1$, and so it is divisorial.

Example 4.9 (Small). Let us start from the $\mathbb{C}^*$-action on the quadric $\mathrm{B}_5(1)$ of Example 4.8 and now consider the induced action on the (18-dimensional) orthogonal Grassmannian

$$X := \mathrm{B}_5(3)$$

of planes in the quadric. It has criticality 2 and three fixed-point components

$$Y_0 \simeq \mathrm{B}_4(2), \quad Y_1 \simeq \mathrm{B}_4(3), \quad Y_2 \simeq \mathrm{B}_4(2).$$

In fact, Y_0 and Y_2 are the subvarieties parametrizing planes of $\mathrm{B}_5(2)$ passing through z_- or z_+, so they are both isomorphic to $\mathrm{B}_4(2)$. The inner one is the subvariety parametrizing planes contained in Z_0, which is isomorphic to $\mathrm{B}_4(3)$.

We claim that the contraction $\pi_1 \colon \mathcal{C}X \to Y_0 \times Y_2$ is small. In fact, we first note that a point in $Y_0 \times Y_2$ corresponds to a pair of lines Λ_0, Λ_2 in $\mathrm{B}_4(1)$. With arguments similar to the ones used in

Example 4.8, we can show that the image of π_1 corresponds to the set of pairs (Λ_0, Λ_2) such that $\Lambda_0 \cap \Lambda_2 \neq \emptyset$, and the image of the exceptional locus corresponds to the set $\{(\Lambda_0, \Lambda_2) |\ \Lambda_0 = \Lambda_2\} \simeq \mathrm{B}_4(2)$, which has dimension 11. The inverse image of each of these elements is 4-dimensional; hence the exceptional locus of π_1 has dimension 15, *i.e.*, codimension 2 in $\mathcal{C}X$, which has dimension $17 = \dim \mathrm{B}_5(3) - 1$.

Let us conclude this section by describing the nef and Mori cones of $\mathcal{C}X$.

Theorem 4.10. *The intersection number $\mathcal{L}_{i,i} \cdot \Gamma_j$ is positive if $i = j$ and zero otherwise. In particular, the numerical class of the curves Γ_j is well defined for every j. Moreover, the nef cone and the Mori cone of $\mathcal{C}X$ are simplicial, generated by the numerical classes of the divisors $\mathcal{L}_{i,i}$, $i \in I$, and of the curves Γ_i, $i \in I$, respectively.*

Proof. By Proposition 4.5, $\mathcal{D}_i \cdot \Gamma_i = 0$ for every $i \in I$; in particular—since every $\mathcal{L}_{j,j}$ is nef—$\mathcal{L}_{j,j} \cdot \Gamma_j = 0$ for $j \neq i$. Since, by Lemma 3.22, $E_j \cdot \Gamma_j = 2$, we see that the numerical class of the curves Γ_j constructed in Construction 3.19 is well defined. Abusing notation, we will write this class as $\Gamma_j \in \mathrm{N}_1(\mathcal{C}X)$.

By [ORSW25, Lemma 6.4], $\mathcal{A} = \mathcal{D}_i \otimes \mathcal{L}_{i,i}$ is ample on $\mathcal{C}X$, so $\mathcal{L}_{i,i} \cdot \Gamma_i > 0$. The set $\{\mathcal{L}_{i,i}\}_{i \in I}$ is linearly independent in $\mathrm{N}^1(\mathcal{C}X)$ by Lemma 4.2. Therefore, the set $\{\Gamma_i\}_{i \in I} \subset \mathrm{N}_1(\mathcal{C}X)$ is also linearly independent, so every class in $\overline{\mathrm{NE}(\mathcal{C}X)}$ can be written as a real linear combination

$$\sum a_i \Gamma_i.$$

Intersecting with $\mathcal{L}_{i,i}$, we get that $a_i \geq 0$ for every $i \in I$, so $\mathrm{NE}(\mathcal{C}X)$ is the cone generated by $\{\Gamma_i\}_{i \in I}$, and by duality, we get the statement on the nef cone. □

Remark 4.11. Let us consider the case in which $\dim Y_0$, $\dim Y_r > 0$. As previously noted, the contractions π_0, π_r are always divisorial, and the contractions π_i, $i = 2, \dots, r-2$, are always small. In the case $r = 2$, we have seen examples (Examples 4.7, 4.8, 4.9) in which π_1 is of fiber type, divisorial, and small. When $r > 2$, the contractions π_1, π_{r-1} are necessarily birational (in fact, the birational maps $\mathcal{C}X \to \mathcal{G}X_{2,2}$, $\mathcal{C}X \to \mathcal{G}X_{r-2,r-2}$ factor via π_1, π_{r-1}, respectively). We have represented in Figure 2 the Mori cones of these Chow quotients in the cases $r = 2, 3, 4$.

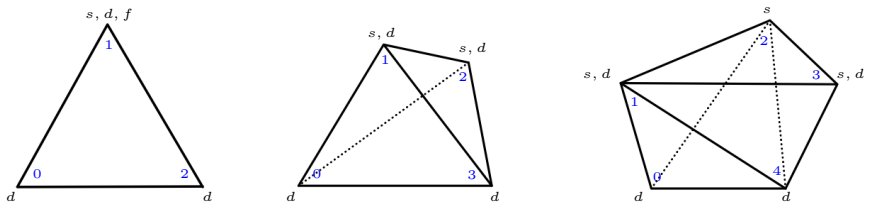


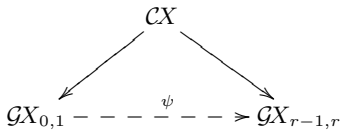
Figure 2: The Mori cone of $\mathcal{C}X$ and the possible types of contractions (d=divisorial, f=fiber type, s=small) in the cases $r = 2, 3, 4$, with no isolated extremal fixed points.

The type (divisorial or small) of the birational contraction π_1 (resp. π_{r-1}) turns out to be determined by the induced action on the varieties $\mathcal{U}_x$, $x \in Y_1$ (respectively, $x \in Y_{r-1}$), which has bandwidth 2. If there is an orbit linking two given elements of the sink and the source of $\mathcal{U}_x$, then the map π_1 (resp. π_{r-1}) is divisorial; if that is not the case, then the map is small. A more detailed study of the (nef, Mori, movable, effective) cones of $\mathcal{C}X$ and of the partial Chow quotients, with special attention to the case in which X is rational homogeneous, will be the goal of a forthcoming paper.

Prior author-source supplement (uncounted)

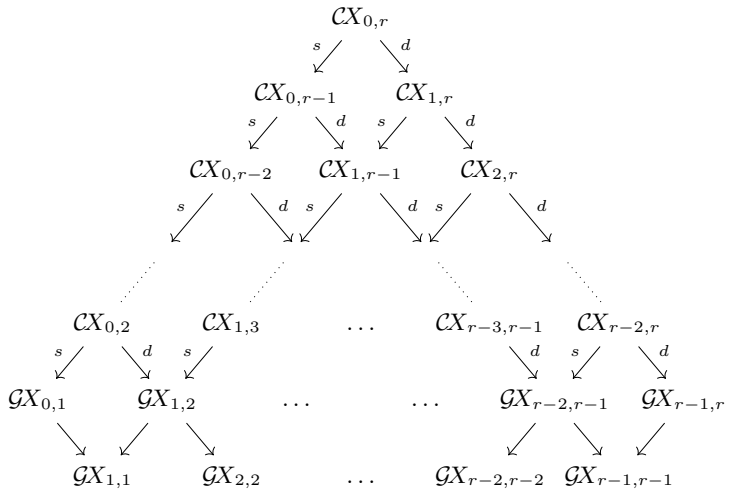
The following separately attributed excerpts are from Occhetta, Romano, Sol'a Conde and Wi'sniewski, arXiv:2310.18623v3 (CC BY 4.0). They supply the earlier Hilbert-scheme argument explicitly invoked by the current article. No result from this prior work is counted here.

We explore here this idea, from the viewpoint of birational geometry. All the geometric quotients of the action are birationally dominated by $\mathcal{C}X$; in particular, we get a resolution of the birational map $\psi : \mathcal{G}X_{0,1} \dashrightarrow \mathcal{G}X_{r-1,r}$:



It then makes sense to study the properties of this resolution and ask ourselves, for instance, whether it gives a Strong Factorization of ψ, that is whether the morphisms $\mathcal{C}X \to \mathcal{G}X_{0,1}$, $\mathcal{C}X \to \mathcal{G}X_{r-1,r}$ are compositions of blowups along smooth centers. What we prove is a structure theorem that shows how to obtain $\mathcal{C}X$ recursively out of the Chow quotients of the birational modifications $X_{i,j}$ described above, via subsequent blowups. More precisely, our main statement is the following (see §2.3 for the definition of $B^{\pm}(Y)$):

Theorem 1.1. *Let (X, L) be a polarized pair, with X smooth, endowed with an equalized nontrivial $\mathbb{C}^*$-action such that* $\operatorname{codim}(B^{\pm}(Y), X) > 1$ *for every inner fixed point component Y. Denote by r be the criticality of the action, by $X_{i,j}$, $0 \le i < j \le r$, the corresponding prunings of X, and by $\mathcal{C}X_{i,j}$ their normalized Chow quotients. Then $\mathcal{C}X = \mathcal{C}X_{0,r}$, and we have a commutative diagram:*



(2)

where all the arrows denoted by s, d are blowups and every rhombus in the diagram is a normalized Cartesian square.

Theorem 1.2. *Under the assumptions of Theorem 1.1, assume moreover that X is convex. Then the varieties $\mathcal{C}X_{i,j}$ with $i = 0$ or $j = r$ are smooth, and the maps*

$$\mathcal{G}X_{0,1} \xleftarrow{s} \mathcal{C}X_{0,2} \xleftarrow{s} \dots \xleftarrow{s} \mathcal{C}X \xrightarrow{d} \dots \xrightarrow{d} \mathcal{C}X_{r-2,r} \xrightarrow{d} \mathcal{G}X_{r-1,r}$$

are smooth blowups, that is, we have a strong factorization of ψ.

2.3 Geometric quotients

It is a well known fact in Mumford's Geometric Invariant Theory that a $\mathbb{C}^*$-action on a polarized variety (X, L) allows to define projective quotients of X, as follows. Let

$$A := \bigoplus_{m \ge 0} \mathrm{H}^0(X, mL)$$

be the *graded $\mathbb{C}$-algebra of sections of L*, and set, for every $\tau \in [0, \delta] \cap \mathbb{Q}$:

$$A(\tau) := \bigoplus_{\substack{m \ge 0 \\ m\tau \in \mathbb{Z}}} \mathrm{H}^0(X, mL)_{m\tau}, \qquad \mathcal{G}X(\tau) := \operatorname{Proj}(A(\tau)).$$

We will call the $\mathcal{G}X(\tau)$'s the *GIT-quotients of the $\mathbb{C}^*$-action on* (X, L).

The above construction has been extended by Białynicki-Birula and Swięcicka to the construction of other (in general non projective) quotients of X, described in terms of semi-sections and sections of the $\mathbb{C}^*$-action (cf. [7]; see also [32]). Their construction allows us to describe easily the open subsets $X^{\mathrm{ss}}(\tau) \subset X$ of semistable points on which the rational maps

$$X \dashrightarrow \mathcal{G}X(\tau)$$

are defined. In order to do so we need the language of *BB-cells*, provided by the *Białynicki-Birula decomposition* of X (see [5, 8]): given a fixed point component $Y \in \mathcal{Y}$, we denote by

$$X^{\pm}(Y) := \{x \in X | \lim_{t^{\pm 1} \to 0} tx \in Y\}, \qquad B^{\pm}(Y) := \overline{X^{\pm}(Y)}, \tag{3}$$

the *BB-cells* of the action and their closures. The equalities:

$$X = \bigsqcup_{Y \in \mathcal{Y}} X^{+}(Y) = \bigsqcup_{Y \in \mathcal{Y}} X^{-}(Y)$$

are called the $+$ and $-$ *BB-decomposition* of X.

To describe $X^{\mathrm{ss}}(\tau)$ for a $\mathbb{C}^*$-action on a polarized pair (X, L), let us define

$$B_i^{\pm} := \bigcup_{\mu_L(Y) = a_i} B^{\pm}(Y)$$

and, for notational reasons, $B_i^{\pm} = \emptyset$ for $i \in \mathbb{Z} \setminus \{0, \ldots, r\}$. Then, following [7], we may write:

$$X^{\mathrm{ss}}(\tau) = X \setminus \Big(\bigcup_{a_i < \tau} B_i^{+} \cup \bigcup_{a_i > \tau} B_i^{-} \Big).$$

In particular, $X^{\mathrm{ss}}(\tau)$ is the same for any $\tau \in (a_i, a_{i+1}) \cap \mathbb{Q}$, and so we set:

- $\mathcal{G}X_{i,i} := \mathcal{G}X(\tau)$ for $\tau = a_i$;
- $\mathcal{G}X_{i,i+1} := \mathcal{G}X(\tau)$ for $\tau \in (a_i, a_{i+1}) \cap \mathbb{Q}$.

Note that $\mathcal{G}X_{i,i+1}$ is a geometric quotient of X for every i, whereas $\mathcal{G}X_{i,i}$ is only semigeometric. Moreover, by construction,

$$\mathcal{G}X_{0,0} = Y_0, \quad \mathcal{G}X_{r,r} = Y_r.$$

Furthermore, since the intersection of all the open sets $X^{\mathrm{ss}}(i, i+1)$ is nonempty, the varieties $\mathcal{G}X(i, i+1)$ are birationally equivalent. The natural birational maps among them fit in the following commutative diagram, whose diagonal arrows are contractions:

$$\begin{array}{ccccccccc} & \mathcal{G}X_{0,1} & \dashrightarrow & \mathcal{G}X_{1,2} & \ldots & \mathcal{G}X_{r-2,r-1} & \dashrightarrow & \mathcal{G}X_{r-1,r} & \\ \swarrow & & \searrow \quad \swarrow & & & & \searrow \quad \swarrow & & \searrow \\ \mathcal{G}X_{0,0} & & \mathcal{G}X_{1,1} & \ldots & & \ldots & \mathcal{G}X_{r-1,r-1} & & \mathcal{G}X_{r,r} \end{array} \tag{4}$$

The horizontal maps are Atiyah flips; this is a known fact which goes back to the work of Thaddeus, Reid and Włodarczyk ([41, 38, 36]).

Lemma 2.6 (AM vs. FM)**.** *Let (X, L) be a polarized pair, with X smooth, supporting an equalized $\mathbb{C}^*$-action, and let C be the closure of a 1-dimensional orbit, whose sink and source are denoted by x_- and x_+. Then C is a smooth rational curve of L-degree equal to $\mu_L(x_+) - \mu_L(x_-)$.*

In this section we consider a nontrivial $\mathbb{C}^*$-action on a polarized pair (X, L) and we study the Chow quotient of X by $\mathbb{C}^*$, a concept due to Kapranov who, in his seminal work [21], defined it and identified it with a moduli of pointed curves when X is a Grassmannian endowed with an action of a complex torus. Slightly different parameter spaces for $\mathbb{C}^*$-invariant cycles were previously proposed by Białynicki-Birula and Sommese, [6], inspired by a construction by Fujiki (cf. [15]).

Following Kapranov's definition, we may find a nonempty open set $V \subset X$ of points x such that the cycles $\overline{\mathbb{C}^* \cdot x}$ belong to a single homology class in X. By shrinking V, if necessary, the geometric quotient of V by the action of $\mathbb{C}^*$ exists. Moreover, taking closures in X of the orbits of points of V, we obtain an algebraic irreducible family of rational curves in X, therefore a morphism $\psi : V/\mathbb{C}^* \to \mathrm{Chow}(X)$.

Definition 4.1. With the above notation, the *Chow quotient* of X by the action of $\mathbb{C}^*$ is defined as the closure:

$$\overline{\mathcal{C}X} := \overline{\psi(V/\mathbb{C}^*)} \subset \mathrm{Chow}(X).$$

We will denote by $\mathcal{C}X$ the normalization of $\overline{\mathcal{C}X}$, and call it the *normalized Chow quotient* of X by the $\mathbb{C}^*$-action. The pullback to $\mathcal{C}X$ of the universal family of $\mathrm{Chow}(X)$ will be denoted by $p : \mathcal{U} \to \mathcal{C}X$, with evaluation morphism $q : \mathcal{U} \to X$.

The $\mathbb{C}^*$-action on X extends naturally to an action on $\mathrm{Chow}(X)$ and the corresponding universal family of cycles. This action restricts to the trivial action on $V/\mathbb{C}^*$, hence to the trivial action on $\mathcal{C}X$. Consequently we have an action on $\mathcal{U}$, so that the maps $p : \mathcal{U} \to \mathcal{C}X$, $q : \mathcal{U} \to X$ are $\mathbb{C}^*$-equivariant, and every cycle parametrized by $\mathcal{C}X$ is $\mathbb{C}^*$-invariant.

Throughout the rest of the paper we will assume that the $\mathbb{C}^*$-action is equalized and that X is smooth. Let us start by observing that in this case every $\mathbb{C}^*$-invariant cycle parametrized by $\mathcal{C}X$ is a chain of rational curves with simple normal crossings (see [27, Lemma 3.12] for a similar statement for the parameter space of stable maps):

Lemma 4.3. *Let (X, L) be a polarized pair, with X smooth, admitting an equalized $\mathbb{C}^*$-action. Let $z \in \mathcal{C}X$ be any point, and let $Z = p^{-1}(z)$ be the corresponding cycle. Then Z is reduced, and its dual graph is of type* A*.*

Proof. Note that Z is necessarily a connected cycle meeting the sink and the source, of L-degree equal to the bandwidth $\delta = a_r$ of the action, by Lemma 2.6. If Z_j is an irreducible component of Z, denoting by z_j^-, z_j^+ its sink and source, the equalization assumption implies that the L-degree of the reduced scheme of Z_j is equal to $\mu_L(z_j^+) - \mu_L(z_j^-)$ (again by Lemma 2.6), and this number is of type $a_i - a_j$ for some $i, j \in \{0, \dots, r\}$. We conclude by summing on the components of Z. □

Remark 4.4. Note that the proof of Lemma 4.3 also implies that an irreducible component Z_k of $Z = p^{-1}(z)$ is a rational curve of degree $\mu_L(z_k^+) - \mu_L(z_k^-)$, where z_k^-, z_k^+ denote the sink and the source of Z_k, respectively. In particular, there are no components of Z contained in a fixed-point component.

A consequence of the reducedness given by Lemma 4.3 is the following statement, which can be obtained by applying standard arguments on parameter spaces:

Lemma 4.5. *Let (X, L) be a polarized pair, with X smooth, with an equalized $\mathbb{C}^*$-action. Then the universal family $p : \mathcal{U} \to \mathcal{C}X$ is flat.*

Proof. Let $Z = p^{-1}(z)$ be any fiber of p. By Lemma 4.3, the components of Z are all reduced, so that Z is normal at the general point of each component and, since $\mathcal{C}X$ is normal, we may apply [23, Theorem I.6.6] to claim that p is flat at the general point of any component of Z. We then conclude by applying [23, Theorem I.7.3.1]. □

Remark 4.6. As a consequence of the flatness of $\mathcal{U} \to \mathcal{C}X$, we obtain a morphism $\mathcal{C}X \to \mathrm{Hilb}(X)$ such that $\mathcal{U}$ is the pullback to $\mathcal{C}X$ of the universal family over $\mathrm{Hilb}(X)$. By shrinking the open subset V introduced above, we can assume that the image of $\mathcal{C}X \to \mathrm{Hilb}(X)$ is the closure of the image of the natural map from $V/\mathbb{C}^*$ to $\mathrm{Hilb}(X)$, whose normalization $\mathcal{H}X$ we call the *Hilbert quotient* of the action. On the other hand we have a natural morphism from $\mathrm{Hilb}(X)$ to $\mathrm{Chow}(X)$ (cf. [23, Theorem I.6.3]), so it turns out that $\mathcal{C}X$ coincides with the Hilbert quotient of X. The idea of considering a flat family of $\mathbb{C}^*$-invariant cycles, whose general element is the closure of a general orbit –in the complex geometric setting– goes back to Fujiki, [15] (see also [6, Theorem 0.1.2]).

Remark 4.7. Given an equalized $\mathbb{C}^*$-action on X, every cycle parametrized by $\mathcal{C}X$ has a unique lift-up to the blowup $X^\flat$ of X along the sink and the source. In particular $\mathcal{C}X \simeq \mathcal{C}X^\flat$.

Proposition 6.4. *Under the hypotheses of Theorem 1.1, assume moreover that X is convex. Then $\mathcal{C}X_{i,j}$ is smooth if $i = 0$ or $j = r$. In particular $\mathcal{C}X$ is smooth.*

Proof. Let us prove the statement for the quotients $\mathcal{C}X_{0,j}$; the case of $\mathcal{C}X_{i,r}$ is analogous. Note that the variety $X_{0,j}$ is by definition the pruning $X(\tau_-, \tau_+)$ for $0 < \tau_- < a_1$, $a_{j-1} < \tau_+ < j$.

Let us consider the blowup of X along the sink Y_0 and the source Y_r of the action, $\beta : X^\flat \to X$, with exceptional divisors $Y_0^\flat$ and $Y_r^\flat$. The indeterminacy locus of the birational modification $X_{0,j} \dashrightarrow X^\flat$ is disjoint from $Y_0^\flat$, so there exists a smooth equivariant blowdown $X_{0,j} \to X'_{0,j}$ which contracts $Y_0^\flat$ to Y_0. Alternatively, one may describe $X'_{0,j}$ as the pruning $X(0, \tau_+)$. The Chow quotients $\mathcal{C}X_{0,j}$ and $\mathcal{C}X'_{0,j}$ are isomorphic (Cf. Remark 4.7), so we will show that $\mathcal{C}X'_{0,j}$ is smooth. To simplify notation, for the rest of the proof we will set $X' := X'_{0,j}$. The sink of the $\mathbb{C}^*$-action on X' is (isomorphic to) Y_0, and the source Y'_r is a divisor which is a birational modification of $Y_r^\flat$.

Let C be the closure of a 1-dimensional orbit of the action with sink and source $x_\pm$, and assume that $x_+ \in Y'_r$. Then the weights of the $\mathbb{C}^*$-action on T_{X',x_+} (respectively T_{X',x_-}) are all 0's and 1's (respectively 1's, 0's and (-1)'s). Then by [33, Lemma 2.16] we get that the degrees of the direct summands of $T_{X'|C}$ cannot be smaller than -1; thus the same holds for $\mathcal{N}_{C/X'}$, and we may conclude that

$$\mathrm{H}^1(C, T_{X'|C}) = \mathrm{H}^1(C, \mathcal{N}_{C/X'}) = 0. \tag{14}$$

Let us now denote by C_g the closure of the general orbit of the action in X', and by $[C_g]$ the corresponding point in $\mathrm{Hilb}(X')$; applying the above argument to C_g we get that $\mathrm{Hilb}(X')$ is smooth at $[C_g]$. In particular there exists a unique irreducible component of $\mathrm{Hilb}(X')$ containing it, that we denote here by $\mathcal{M}$. Note that $\mathcal{M}$ is a projective variety. Let us now consider the induced $\mathbb{C}^*$-action on $\mathcal{M}$; since $\mathcal{M}$ is smooth at $[C_g]$, by [5, Theorem 2.5] there exists a unique fixed point component $H \subset \mathcal{M}$ containing $[C_g]$. We will show that $\mathcal{M}$ is smooth at every point of H, from which the smoothness of H will follow (again by [5, Theorem 2.5]).

By Remark 4.6, the normalization of H is $\mathcal{C}X'$; in particular, for every $[C] \in H$, the corresponding subscheme C is a reduced tree of smooth rational curves (Lemma 4.3). Furthermore, the intersection of two meeting components of C is transversal, since at every fixed point x' the positive and the negative part of $T_{X',x'}$ are transversal. In particular, the subscheme C is a local complete intersection, hence the conormal sheaf $\mathcal{I}_{C,X'}/\mathcal{I}^2_{C,X'}$ is locally free, and we have a short exact sequence:

$$0 \longrightarrow \mathcal{I}_{C,X'}/\mathcal{I}^2_{C,X'} \longrightarrow \Omega_{X'|C} \longrightarrow \Omega_C \longrightarrow 0$$

By applying the functor $\mathrm{Hom}(_, \mathcal{O}_C)$, we get that $\mathrm{Ext}^1(\mathcal{I}_{C,X'}/\mathcal{I}^2_{C,X'}, \mathcal{O}_C)$ is a quotient of $\mathrm{Ext}^1(\Omega_{X'|C}, \mathcal{O}_C) = \mathrm{H}^1(C, T_{X'|C})$. To conclude that H is smooth at $[C]$ it is enough to show (see, for instance, [23, Theorem I.2.8]) that

$$\mathrm{H}^1(C, T_{X'|C}) = 0. \tag{15}$$

To prove (15) we write C as $C_1 \cup C_2$, with C_2 irreducible and meeting the source, and set $\{p\} = C_1 \cap C_2$. By (14), we know that $\mathrm{H}^1(C, T_{X'}|_{C_2}) = 0$, and so if $C = C_2$ the proof is finished. If else $C_1 \neq \emptyset$ we consider the exact sequence

$$0 \longrightarrow T_{X'}|_{C_1}(-p) \longrightarrow T_{X'}|_C \longrightarrow T_{X'}|_{C_2} \longrightarrow 0$$

Since X' is isomorphic to X in a neighborhood of C_1, and X is convex by assumption, we have $\mathrm{H}^1(C_1, T_{X'}|_{C_1}(-p)) = 0$ by (32) in the Proof of [16, Lemma 10], so (15) holds. □

Corollary 7.1. *Let (X, L) be a polarized pair, with X smooth, endowed with an equalized nontrivial $\mathbb{C}^*$-action. Then the conclusions of Theorems 1.1 and 1.2 hold if the action satisfies the following condition:*

$$\mathrm{codim}(B^\pm(Y), X) > 1 \text{ for every } Y \in \mathcal{Y} \text{ of weight } a_i \text{ with } 2 \leq i \leq r-2. \tag{$\star$}$$

Lemma 7.4. *Let (X, L) be a polarized pair, with X smooth, endowed with an equalized nontrivial $\mathbb{C}^*$-action, satisfying Condition ($\star$). Let $\mathcal{L}_{i,i+1} \in \mathrm{Pic}(\mathcal{C}X)$ (resp. $\mathcal{L}_{i,i} \in \mathrm{Pic}(\mathcal{C}X)$) be pullbacks of ample line bundles on $\mathcal{G}X_{i,i+1}$, (resp. on $\mathcal{G}X_{i,i}$). Then the line bundles*

$$\mathcal{L}_{0,1} \otimes \cdots \otimes \mathcal{L}_{r-1,r}, \qquad \mathcal{L}_{0,0} \otimes \cdots \otimes \mathcal{L}_{r,r}$$

are ample on $\mathcal{C}X = \mathcal{C}X_{0,r}$.

Proof. To prove that $\mathcal{L}_{0,1} \otimes \cdots \otimes \mathcal{L}_{r-1,r}$ is ample it is enough to show that the product morphism $\mathcal{C}X_{0,r} \to \Pi_i \mathcal{G}X_{i,i+1}$ is finite. This morphism maps a point $c \in \mathcal{C}X_{0,r}$ parametrizing a cycle C whose irreducible components $C_1, \ldots, C_s$ are closures of orbits, to a point of the form $([C_{j_0}], \ldots, [C_{j_{r-1}}])$, for certain $j_0, \ldots, j_s \in \{0, \ldots, r\}$. By definition C_{j_i} is the (unique) irreducible component of C such that the weights of L at its sink and source are smaller than or equal to a_i, and bigger than or equal to a_{i+1}, respectively.

Therefore two cycles in the inverse image of a point c' in the image of $\mathcal{C}X_{0,r} \to \Pi_i \mathcal{G}X_{i,i+1}$ have the same irreducible components; by Lemma 4.3 they have the same image in the Chow quotient $\overline{\mathcal{C}X} \subset \mathrm{Chow}(X)$. Since $\mathcal{C}X$ is the normalization of $\overline{\mathcal{C}X}$, the result follows.

The ampleness of $\mathcal{L}_{0,0} \otimes \cdots \otimes \mathcal{L}_{r,r}$ will follow from the first part if we show that the product morphisms $\mathcal{G}X_{i,i+1} \to \mathcal{G}X_{i,i} \times \mathcal{G}X_{i+1,i+1}$ are finite, for every i. Let us notice that two elements $[C]$ and $[C']$ of $\mathcal{G}X_{i,i+1}$ are mapped to the same element of $\mathcal{G}X_{i,i}$ (resp. $\mathcal{G}X_{i+1,i+1}$) if and only if they have the same sink, with L-weight a_i (resp. the same source, with L-weight a_{i+1}). Assume by contradiction that for some i there is a curve $B \subset \mathcal{G}X_{i+i+1}$ which is contracted by $\mathcal{G}X_{i,i+1} \to \mathcal{G}X_{i,i} \times \mathcal{G}X_{i+1,i+1}$. Then there is a one dimensional family $\{C_b, b \in B\}$ of closures of orbits with the same sink of L-weight a_i and the same source of L-weight a_{i+1}, hence, by Mori's Bend-and-Break (see [11, Proposition 3.2]) a reducible or non reduced $\mathbb{C}^*$-invariant cycle with sink of L-weight a_i and source of L-weight a_{i+1}, a contradiction. □

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